

Norway

SLEIPNER

Volve F

F-15

F-15A

Design: F-15A

Standard Survey Report

11 april, 2018

Statoil
Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Site:	Volve F	MD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Well:	F-15	North Reference:	Grid
Wellbore:	F-15A	Survey Calculation Method:	Minimum Curvature
Design:	F-15A	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude: 58° 26' 29.8068 N
From: Map		Easting:	435,050.02 m	Longitude: 1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence: -0.95 °

Well	F-15			
Well Position	+N/-S	-3.17 m	Northing:	6,478,560.36 m
	+E/-W	3.53 m	Easting:	435,053.55 m
Position Uncertainty		0.00 m	Wellhead Depth:	91.00 m
			Water Depth:	91.00 m

Wellbore	F-15A				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	BGGM2009	31.12.2008	-2.07	71.61	50,390

Design	F-15A				
Audit Notes:	Bit TD changed from 4095 mMD to 4097 mMD according to DBR.				
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	1,381.00
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	145.90	-3.17	3.53	247.56	

Survey Program	Date	06.08.2012			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
153.40	1,364.50	12 1/4" F-15 Drop Gyro (F-15)	Wellbore Surveyor, stat	Gyro Tool from GD	
1,383.00	2,513.00	12 1/4" Drop Gyro F-15A Revised (F-15A)	Wellbore Surveyor, stat	Gyro Tool from GD	
2,544.30	4,018.65	8 1/2" Drop gyro F-15A (F-15A)	Wellbore Surveyor, stat	Gyro Tool from GD	
3,199.80	3,684.07	8 1/2" MWD F15A (F-15A)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)	
3,713.92	4,018.65	8 1/2" Drop gyro F-15A (F-15A)	Wellbore Surveyor, stat	Gyro Tool from GD	

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	-3.17	3.53	0.00	0.000	0.000	0.000	
153.40	0.15	253.69	153.40	-3.17	3.52	0.01	0.600	0.600	0.000	
234.20	0.28	146.02	234.20	-3.37	3.53	0.08	0.132	0.048	-39.976	
274.60	0.78	136.21	274.60	-3.65	3.78	-0.04	0.376	0.371	-7.285	
315.00	1.46	128.29	314.99	-4.16	4.37	-0.40	0.517	0.505	-5.881	
355.30	3.41	124.07	355.25	-5.15	5.77	-1.31	1.457	1.452	-3.141	
395.70	5.13	126.98	395.54	-6.91	8.21	-2.89	1.287	1.277	2.161	

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Site:	Volve F	MD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Well:	F-15	North Reference:	Grid
Wellbore:	F-15A	Survey Calculation Method:	Minimum Curvature
Design:	F-15A	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
436.10	5.52	139.07	435.76	-9.47	10.92	-4.43	0.880	0.290	8.978	
474.40	5.62	152.33	473.89	-12.52	13.00	-5.18	1.009	0.078	10.386	
514.80	6.19	163.05	514.07	-16.36	14.55	-5.15	0.920	0.423	7.960	
557.10	7.03	171.79	556.09	-21.10	15.59	-4.30	0.928	0.596	6.199	
597.50	7.74	174.05	596.16	-26.25	16.22	-2.92	0.570	0.527	1.678	
637.80	8.49	174.54	636.05	-31.91	16.79	-1.28	0.561	0.558	0.365	
678.20	8.94	175.57	675.99	-38.01	17.31	0.56	0.354	0.334	0.765	
718.50	9.34	177.45	715.77	-44.40	17.70	2.64	0.372	0.298	1.400	
758.90	10.22	185.51	755.59	-51.24	17.50	5.44	1.207	0.653	5.985	
799.30	11.89	193.59	795.24	-58.86	16.18	9.57	1.689	1.240	6.000	
839.70	13.68	199.13	834.64	-67.42	13.64	15.19	1.610	1.329	4.114	
880.00	15.14	201.54	873.67	-76.81	10.14	22.00	1.175	1.087	1.794	
920.40	13.72	203.30	912.79	-86.12	6.31	29.10	1.103	-1.054	1.307	
960.80	14.53	201.31	951.97	-95.24	2.57	36.04	0.701	0.601	-1.478	
1,001.10	16.51	200.62	990.80	-105.31	-1.28	43.44	1.480	1.474	-0.514	
1,041.60	16.33	201.19	1,029.65	-116.01	-5.37	51.30	0.179	-0.133	0.422	
1,081.90	16.18	205.25	1,068.34	-126.37	-9.81	59.36	0.853	-0.112	3.022	
1,122.30	17.54	211.88	1,107.01	-136.63	-15.43	68.47	1.747	1.010	4.923	
1,162.60	17.66	221.21	1,145.43	-146.39	-22.66	78.88	2.100	0.089	6.945	
1,203.00	16.05	230.28	1,184.10	-154.57	-31.00	89.71	2.286	-1.196	6.735	
1,243.30	15.33	232.77	1,222.90	-161.35	-39.52	100.18	0.734	-0.536	1.854	
1,283.80	18.17	237.03	1,261.68	-168.03	-49.09	111.57	2.291	2.104	3.156	
1,324.10	21.21	241.08	1,299.62	-174.98	-60.74	124.99	2.479	2.263	3.015	
1,364.50	24.71	243.03	1,336.81	-182.34	-74.67	140.68	2.659	2.599	1.448	
1,383.00	26.95	242.95	1,353.46	-186.01	-81.85	148.71	3.633	3.632	-0.130	
1,406.00	25.71	234.70	1,374.08	-191.26	-90.57	158.78	5.035	-1.617	-10.761	
1,445.00	24.44	222.89	1,409.43	-202.07	-102.97	174.36	3.966	-0.977	-9.085	
1,487.00	21.59	210.72	1,448.10	-215.09	-112.84	188.46	3.952	-2.036	-8.693	
1,544.00	20.56	192.78	1,501.35	-233.88	-120.42	202.64	3.425	-0.542	-9.442	
1,584.00	21.08	181.10	1,538.75	-247.93	-122.11	209.56	3.133	0.390	-8.760	
1,625.00	21.07	170.97	1,577.02	-262.59	-121.10	214.22	2.662	-0.007	-7.412	
1,665.00	20.92	159.85	1,614.38	-276.40	-117.51	216.17	2.986	-0.112	-8.340	
1,705.00	22.16	151.36	1,651.59	-289.73	-111.43	215.64	2.513	0.930	-6.367	
1,746.00	24.79	147.70	1,689.20	-303.78	-103.13	213.34	2.199	1.924	-2.678	
1,786.00	27.24	144.45	1,725.15	-318.32	-93.32	209.82	2.125	1.837	-2.437	
1,827.00	27.70	143.46	1,761.52	-333.61	-82.19	205.38	0.474	0.337	-0.724	
1,867.00	28.14	143.47	1,796.87	-348.66	-71.04	200.81	0.330	0.330	0.007	
1,907.00	27.08	143.10	1,832.31	-363.52	-59.96	196.25	0.805	-0.795	-0.277	
1,948.00	27.31	143.53	1,868.78	-378.55	-48.77	191.64	0.221	0.168	0.315	
1,988.00	27.71	143.72	1,904.26	-393.42	-37.81	187.19	0.307	0.300	0.142	
2,028.00	27.86	145.00	1,939.64	-408.58	-26.95	182.93	0.461	0.112	0.960	
2,069.00	27.42	149.82	1,975.97	-424.59	-16.70	179.58	1.667	-0.322	3.527	
2,109.00	26.74	154.96	2,011.59	-440.70	-8.26	177.93	1.827	-0.510	3.855	

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Site:	Volve F	MD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Well:	F-15	North Reference:	Grid
Wellbore:	F-15A	Survey Calculation Method:	Minimum Curvature
Design:	F-15A	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
2,149.00	25.09	162.89	2,047.58	-456.97	-1.96	178.31	2.876	-1.237	5.947	
2,190.00	24.05	170.43	2,084.87	-473.52	1.99	180.98	2.415	-0.761	5.517	
2,230.00	24.40	174.59	2,121.35	-489.78	4.13	185.21	1.307	0.262	3.120	
2,270.00	23.87	181.76	2,157.87	-506.10	4.66	190.95	2.233	-0.397	5.377	
2,311.00	24.72	188.97	2,195.24	-522.86	3.06	198.82	2.256	0.622	5.276	
2,351.00	25.05	194.20	2,231.53	-539.34	-0.32	208.24	1.669	0.247	3.922	
2,392.00	25.95	202.77	2,268.55	-556.03	-5.92	219.79	2.776	0.659	6.271	
2,432.00	27.22	209.99	2,304.33	-572.03	-13.88	233.25	2.602	0.952	5.415	
2,486.00	28.73	219.32	2,352.04	-592.78	-28.29	254.49	2.569	0.839	5.183	
2,513.00	30.35	223.67	2,375.54	-602.73	-37.11	266.44	2.985	1.800	4.833	
2,544.30	32.21	227.05	2,402.29	-614.14	-48.68	281.49	2.450	1.783	3.240	
2,584.60	36.06	231.49	2,435.64	-628.85	-65.83	302.96	3.412	2.866	3.305	
2,624.91	39.18	235.66	2,467.57	-643.42	-85.64	326.83	2.996	2.322	3.103	
2,665.26	41.67	241.21	2,498.30	-657.08	-107.93	352.64	3.252	1.851	4.126	
2,705.63	43.76	244.74	2,527.96	-669.50	-132.32	379.93	2.361	1.553	2.623	
2,745.99	46.14	249.81	2,556.53	-680.49	-158.61	408.42	3.196	1.769	3.769	
2,786.36	48.79	250.79	2,583.82	-690.51	-186.62	438.13	2.041	1.969	0.728	
2,826.70	51.38	251.73	2,609.70	-700.44	-215.92	469.01	1.999	1.926	0.699	
2,867.05	51.74	251.02	2,634.79	-710.54	-245.86	500.54	0.493	0.268	-0.528	
2,907.38	49.19	251.75	2,660.46	-720.47	-275.34	531.57	1.943	-1.897	0.543	
2,947.73	49.52	257.15	2,686.75	-728.67	-304.81	561.95	3.056	0.245	4.015	
2,988.07	50.68	263.14	2,712.64	-733.95	-335.28	592.12	3.524	0.863	4.455	
3,028.40	52.73	265.32	2,737.64	-737.12	-366.77	622.44	1.986	1.525	1.622	
3,068.70	57.54	267.07	2,760.67	-739.30	-399.75	653.75	3.737	3.581	1.303	
3,109.03	59.86	268.74	2,781.62	-740.55	-434.18	686.06	2.026	1.726	1.242	
3,149.35	60.21	270.16	2,801.76	-740.89	-469.11	718.47	0.952	0.260	1.057	
3,189.70	59.32	272.37	2,822.08	-740.12	-503.96	750.38	1.566	-0.662	1.643	
3,199.80	61.42	272.26	2,827.07	-739.76	-512.73	758.36	6.244	6.238	-0.327	
3,241.65	67.42	275.37	2,845.14	-737.23	-550.37	792.18	4.747	4.301	2.229	
3,280.66	74.58	277.80	2,857.83	-732.99	-586.98	824.40	5.782	5.506	1.869	
3,321.38	82.32	278.66	2,865.98	-727.28	-626.44	858.68	5.736	5.702	0.634	
3,361.74	85.13	279.03	2,870.39	-721.11	-666.07	892.96	2.107	2.089	0.275	
3,401.25	85.77	280.03	2,873.52	-714.59	-704.91	926.37	0.899	0.486	0.759	
3,442.62	84.22	280.17	2,877.13	-707.36	-745.49	961.11	1.129	-1.124	0.102	
3,482.50	82.72	281.07	2,881.67	-700.06	-784.43	994.32	1.314	-1.128	0.677	
3,523.20	79.25	283.21	2,888.04	-691.61	-823.72	1,027.41	2.995	-2.558	1.577	
3,563.45	75.59	285.71	2,896.81	-681.81	-861.75	1,058.82	3.278	-2.728	1.863	
3,604.04	71.94	287.70	2,908.16	-670.62	-899.07	1,089.04	3.045	-2.698	1.471	
3,644.20	68.44	290.44	2,921.77	-658.28	-934.77	1,117.33	3.247	-2.615	2.047	
3,684.07	62.69	290.52	2,938.25	-645.59	-968.76	1,143.90	4.327	-4.327	0.060	
3,713.92	59.50	289.88	2,952.68	-636.56	-993.28	1,163.12	3.255	-3.206	-0.643	
3,754.26	55.27	290.09	2,974.42	-624.96	-1,025.21	1,188.20	3.149	-3.146	0.156	
3,794.64	52.20	291.64	2,998.30	-613.37	-1,055.63	1,211.89	2.462	-2.281	1.152	
3,834.97	49.43	295.43	3,023.79	-600.91	-1,084.29	1,233.62	3.003	-2.061	2.819	

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Site:	Volve F	MD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Well:	F-15	North Reference:	Grid
Wellbore:	F-15A	Survey Calculation Method:	Minimum Curvature
Design:	F-15A	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
3,875.29	47.26	297.94	3,050.59	-587.40	-1,111.20	1,253.34	2.134	-1.615	1.868	
3,915.58	45.75	297.73	3,078.32	-573.75	-1,137.05	1,272.02	1.130	-1.124	-0.156	
3,955.87	42.94	297.66	3,107.13	-560.66	-1,161.98	1,290.06	2.093	-2.092	-0.052	
3,996.20	41.91	298.05	3,136.90	-547.95	-1,186.04	1,307.45	0.791	-0.766	0.290	
4,018.65	40.20	298.08	3,153.83	-541.01	-1,199.05	1,316.82	2.285	-2.285	0.040	
4,095.00	40.20	298.08	3,212.14	-517.82	-1,242.53	1,348.15	0.000	0.000	0.000	

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
220.60	220.60	30"	30.000	36.000	
1,368.40	1,340.35	20"	20.000	26.000	
2,561.80	2,416.96	13 3/8"	13.375	16.000	

Checked By: _____	Approved By: _____	Date: _____
-------------------	--------------------	-------------